

[Aim: 100/100 in Maths]

आरंभ

TRIGONOMETRY

Lecture #7

#A7K²:

$$\left[\begin{array}{l} \sin^2 \theta + \cos^2 \theta = 1 \\ \sec^2 \theta - \tan^2 \theta = 1 \\ \operatorname{cosec}^2 \theta - \cot^2 \theta = 1 \end{array} \right]$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin \leftrightarrow \frac{1}{\operatorname{cosec}}$$

$$\cos \leftrightarrow \frac{1}{\sec}$$

$$\tan \leftrightarrow \frac{1}{\cot}$$

$$\underline{\sec \theta + \tan \theta} \leftrightarrow \frac{1}{\underline{\sec \theta - \tan \theta}}$$

$$\operatorname{cosec} \theta - \cot \theta \leftrightarrow \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

#LP: $\frac{1}{\sec A + \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A - \tan A}$ (CBSE 2005)

LHS $(\cancel{\sec A} - \tan A) - \cancel{\sec A}$ RHS

$$\boxed{LHS = -\tan A}$$

$$\begin{aligned} \sec A - (\sec A + \tan A) \\ \cancel{\sec A} - \cancel{\sec A} - \tan A \end{aligned}$$

$$\boxed{RHS = -\tan A}$$

Since $LHS = RHS$

~~HP~~

#LP: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

$$\Rightarrow (\sin^2 A + \operatorname{cosec}^2 A + 2 \sin A \operatorname{cosec} A) + (\cos^2 A + \sec^2 A + 2 \cos A \sec A)$$

$\left[\begin{array}{l} \because \operatorname{cosec} A = \frac{1}{\sin A} \\ \sec A = \frac{1}{\cos A} \end{array} \right]$

$$\Rightarrow \sin^2 A + \operatorname{cosec}^2 A + 2 + \cos^2 A + \sec^2 A + 2$$

$$\Rightarrow 2 + 1 + 1 + \cot^2 A + 1 + \tan^2 A$$

$$\Rightarrow 7 + \cot^2 A + \tan^2 A \Rightarrow \text{RHS}$$

HP

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

#LP: $\frac{\cos^2 \theta}{\sin^2 \theta} - \frac{1}{\sin^2 \theta}$ is equal to

(A) $\tan^2 \theta$

(B) $\sec^2 \theta$

(C) 1

~~(D) -1~~

$$\cot^2 \theta - \csc^2 \theta$$

$$\Rightarrow -(\csc^2 \theta - \cot^2 \theta)$$

$$\Rightarrow - (1)$$

[CBSE 2023]

$$\csc^2 \theta - \cot^2 \theta = 1$$

#LP: The value of $\tan^2\theta - \left(\frac{1}{\cos\theta} \times \sec\theta \right)$ is: [CBSE 2025]

(A) 1

(B) 0

✓ (C) -1

(D) 2

$$\tan^2\theta - \left(\sec\theta \times \sec\theta \right)$$

$$\rightarrow \tan^2\theta - \sec^2\theta$$

$$\rightarrow - \left(\sec^2\theta - \tan^2\theta \right)$$

$$\rightarrow -1$$

$$\sec^2\theta - \tan^2\theta = 1$$

#LP: $(\cos^4 A - \sin^4 A)$ is equal to

[CBSE 2023]

(A) ~~$2 \sin^2 A - 1$~~

(B) ~~$2 \sin^2 A + 1$~~

(C) $2 \cos^2 A + 1$

(D) $2 \cos^2 A - 1$

$$\Rightarrow (\cos^2 A)^2 - (\sin^2 A)^2$$

$$\Rightarrow \underbrace{(\cos^2 A + \sin^2 A)}_{1 \times} \underbrace{(\cos^2 A - \sin^2 A)}_{(\cos^2 A - (1 - \cos^2 A))}$$

$$\Rightarrow \cos^2 A - 1 + \cos^2 A$$

$$\Rightarrow \boxed{2 \cos^2 A - 1}$$

#LP: If $x = a \cos \theta$ and $y = b \sin \theta$, then $b^2 x^2 + a^2 y^2 =$

☒ (A) $a^2 b^2$

☐ (B) ab

☐ (C) $a^4 b^4$

☐ (D) $a^2 + b^2$

$$\Rightarrow b^2 (a \cos \theta)^2 + a^2 (b \sin \theta)^2$$

$$\Rightarrow \underline{b^2 a^2 \cos^2 \theta} + \underline{a^2 b^2 \sin^2 \theta}$$

$$\Rightarrow a^2 b^2 [\cos^2 \theta + \sin^2 \theta]$$

$$\Rightarrow a^2 b^2 \times 1$$

$$\Rightarrow \underline{\underline{a^2 b^2}}$$

Rationalise

#LP: $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \operatorname{cosec} \theta - \cot \theta$

LHS

$$\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta}}$$

$$\Rightarrow \sqrt{\frac{(1 - \cos \theta)^2}{(1)^2 - \cos^2 \theta}}$$

$$\Rightarrow \sqrt{\frac{(1 - \cos \theta)^2}{\sin^2 \theta}} \quad (\because)$$

~~$$\sqrt{\left(\frac{1 - \cos \theta}{\sin \theta}\right)^2}$$~~

$$\Rightarrow \frac{1 - \cos \theta}{\sin \theta}$$

$$\Rightarrow \frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}$$

$$\Rightarrow \boxed{\operatorname{cosec} \theta - \cot \theta}$$

~~HP~~

#LP: Prove that:

(i) $\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} + \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} = 2 \operatorname{cosec} \theta$ (CBSE 2001, 2006)

LHS

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1} \times \frac{\sec \theta - 1}{\sec \theta - 1}} + \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1} \times \frac{\sec \theta + 1}{\sec \theta + 1}}$$

$$\Rightarrow \sqrt{\frac{(\sec \theta - 1)^2}{\sec^2 \theta - 1^2}} + \sqrt{\frac{(\sec \theta + 1)^2}{\sec^2 \theta - 1^2}}$$

$$\Rightarrow \sqrt{\frac{(\sec \theta - 1)^2}{\tan^2 \theta}} + \sqrt{\frac{(\sec \theta + 1)^2}{\tan^2 \theta}}$$

$$\Rightarrow \sqrt{\left(\frac{\sec \theta - 1}{\tan \theta}\right)^2} + \sqrt{\left(\frac{\sec \theta + 1}{\tan \theta}\right)^2}$$

$$\Rightarrow \frac{\sec \theta - 1}{\tan \theta} + \frac{\sec \theta + 1}{\tan \theta}$$

$$\Rightarrow \frac{\sec \theta - \cancel{1} + \sec \theta + \cancel{1}}{\tan \theta}$$

$$\Rightarrow \frac{2 \sec \theta}{\tan \theta}$$

$$\Rightarrow 2 \times \frac{\cancel{\cos \theta}}{\frac{\sin \theta}{\cancel{\cos \theta}}} \Rightarrow 2 \times \frac{1}{\sin \theta} \Rightarrow \boxed{2 \operatorname{cosec} \theta} \text{ (RHS)}$$

#LP: Prove that $\tan^4 A + \tan^2 A = \sec^4 A - \sec^2 A$

LHS

$$\tan^2 A (\tan^2 A + 1)$$

$$\rightarrow \tan^2 A (\sec^2 A)$$

$$\Rightarrow (\sec^2 A - 1) \sec^2 A$$

$$\rightarrow \sec^4 A - \sec^2 A$$

RP

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\sec^2 \theta - 1 = \tan^2 \theta$$

#LP: Prove that $\operatorname{cosec}^4 A - \operatorname{cosec}^2 A = \cot^4 A + \cot^2 A$

LHS

$$\operatorname{cosec}^4 A (\operatorname{cosec}^2 A - 1)$$

$$\Rightarrow \operatorname{cosec}^2 A (\cot^2 A)$$

$$\Rightarrow (1 + \cot^2 A) (\cot^2 A)$$

$$\Rightarrow \cot^2 A + \cot^4 A \quad \text{H/P}$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta$$

$$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

#LP: Prove the following identities: [cbse]

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$\sin \theta = s$$

$$\cos \theta = c$$

$$\frac{\cos^2 \theta}{(1 - \tan \theta)} + \frac{\sin^3 \theta}{(\sin \theta - \cos \theta)} = 1 + \sin \theta \cos \theta$$

LHS

$$\Rightarrow \frac{c^2}{1 - \frac{s}{c}} + \frac{s^3}{s - c}$$

$$\Rightarrow \frac{\frac{c^2}{c - s}}{\frac{c - s}{c}} + \frac{s^3}{s - c}$$

$$\Rightarrow \frac{\frac{c^3}{c - s}}{\frac{s - c}{c - s}}$$

$$\Rightarrow \frac{c^3}{c - s} - \frac{s^3}{c - s}$$

$$\Rightarrow \frac{c^3 - s^3}{c - s}$$

$$\Rightarrow \frac{(c - s)(c^2 + s^2 + c \cdot s)}{c - s}$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta + \cos \theta \sin \theta$$

$$1 + \cos \theta \sin \theta \Rightarrow \text{RHS}$$

QED

#LP: Prove the following identities:

Let $\sin A = s$
 $\cos A = c$

(I)

$$\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$$

(II)

$$= 1 + \tan A + \cot A = 1 + \sec A \operatorname{cosec} A$$

(I) LHS

$$\frac{\frac{s}{c}}{1 - \frac{c}{s}} + \frac{\frac{c}{s}}{1 - \frac{s}{c}}$$

$$\Rightarrow \frac{\frac{s}{c}}{\frac{s-c}{s}} + \frac{\frac{c}{s}}{\frac{c-s}{c}}$$

$$\Rightarrow \frac{s}{c} \times \frac{s}{s-c} + \frac{c}{s} \times \frac{c}{c-s}$$

$$\Rightarrow \frac{s^2}{c(s-c)} + \frac{c^2}{s(c-s)}$$

$$\Rightarrow \frac{s^2}{c(s-c)} - \frac{c^2}{s(s-c)}$$

$$\Rightarrow \frac{1}{(s-c)} \left[\frac{s^2}{c} - \frac{c^2}{s} \right]$$

$$\Rightarrow \frac{1}{(s-c)} \left[\frac{s^2 \cdot s - c^2 \cdot c}{c \cdot s} \right]$$

$$\Rightarrow \frac{1}{(s-c)} \left[\frac{s^3 - c^3}{c \cdot s} \right]$$

$$\frac{1}{\cos A} \quad \frac{1}{\sin A}$$

$$\Rightarrow \frac{1}{(s-c)} \times \frac{(s-c)(s^2 + c^2 + s \cdot c)}{c \cdot s}$$

$$\Rightarrow \frac{1 + s \cdot c}{s \cdot c}$$

$$\Rightarrow \frac{1}{s \cdot c} + \frac{s \cdot c}{s \cdot c}$$

$$\Rightarrow \operatorname{cosec} A \sec A = 1$$

RHS

① LHS

$$1 + \tan A + \cot A$$

$$\rightarrow 1 + \frac{s}{c} + \frac{c}{s}$$

$$\rightarrow \frac{c \cdot s + \sqrt{s^2 + c^2}}{c \cdot s}$$

$$\rightarrow \frac{c \cdot s + 1}{c \cdot s}$$

$$\rightarrow \frac{1}{\cancel{c \cdot s}} + \frac{1}{c \cdot s}$$

$$\rightarrow \boxed{1 + \sec \theta \csc \theta} \Rightarrow \text{RHS} \quad \text{QED}$$

#LP: Prove that: $(\sec\theta - \csc\theta)(1 + \tan\theta + \cot\theta) = \sec\theta\tan\theta - \csc\theta\cot\theta$

LHS $\Rightarrow \sec\theta + \sec\theta\tan\theta + \sec\theta\cot\theta - \csc\theta - \csc\theta\tan\theta - \csc\theta\cot\theta$

$\Rightarrow \sec\theta\tan\theta - \csc\theta\cot\theta + \sec\theta + \frac{1}{\cancel{c}} \times \frac{\cancel{c}}{s} - \csc\theta - \frac{1}{\cancel{s}} \times \frac{\cancel{s}}{c}$

$\Rightarrow \sec\theta\tan\theta - \csc\theta\cot\theta + \cancel{\sec\theta} + \cancel{\csc\theta} - \cancel{\csc\theta} - \cancel{\sec\theta}$

= RHS

LP

#LP: $\frac{(\tan^3 \theta - 1)}{(\tan \theta - 1)} = \sec^2 \theta + \tan \theta$

LHS

$$\frac{\tan^3 \theta - 1^3}{\tan \theta - 1}$$
$$\rightarrow \frac{(\cancel{t-1})(t^2 + 1^2 + t \cdot 1)}{\cancel{t-1}}$$

$$\rightarrow \tan^2 \theta + 1 + \tan \theta$$
$$\rightarrow \boxed{\sec^2 \theta + \tan \theta} \quad \text{HP}$$

= RHS

#LP: Prove the following identities:

$$\frac{(\tan\theta + \sec\theta - 1)}{(\tan\theta - \sec\theta + 1)} = \frac{(1 + \sin\theta)}{\cos\theta}$$

LHS

$$\frac{\tan\theta + \sec\theta - \left(\frac{\sec^2\theta}{a^2} - \frac{\tan^2\theta}{b^2}\right)}{\tan\theta - \sec\theta + 1}$$

$$\left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \sec^2\theta - \tan^2\theta = 1 \right]$$

$$\rightarrow \frac{(\tan\theta + \sec\theta) - \frac{a+b}{a^2} (\sec\theta + \tan\theta) (\sec\theta - \tan\theta)}{\tan\theta - \sec\theta + 1}$$

$$\Rightarrow \frac{(\tan\theta + \sec\theta) [1 - \sec\theta + \tan\theta]}{[\tan\theta - \sec\theta + 1]}$$

$$\rightarrow \tan\theta + \sec\theta$$

$$\rightarrow \frac{\frac{s}{c} + \frac{1}{c}}{\frac{1 + \sin\theta}{\cos\theta}} = \text{RHS}$$

~~QED~~

#LP: Prove the following identities:

$$\frac{\cot A + \operatorname{cosec} A - 1}{\cot A - \operatorname{cosec} A + 1} = \frac{1 + \cos A}{\sin A}$$

(LHS)

$$\Rightarrow \frac{\cot A + \operatorname{cosec} A - (\operatorname{cosec}^2 A - \cot^2 A)}{\cot A - \operatorname{cosec} A + 1}$$

$$\Rightarrow \frac{(\cot A + \operatorname{cosec} A) - (\operatorname{cosec}^2 A - \cot^2 A)}{\cot A - \operatorname{cosec} A + 1}$$

$$\Rightarrow (\cot A + \operatorname{cosec} A) [1 - \operatorname{cosec} A + \cot A]$$

$$\cot A - \operatorname{cosec} A + 1$$

$$\Rightarrow \cot A + \operatorname{cosec} A$$

$$\Rightarrow \frac{\cos A}{\sin A} + \frac{1}{\sin A}$$

$$\Rightarrow \frac{\cos A + 1}{\sin A} = \text{RHS}$$

(HP)

#LP: Prove the following identities:

$$\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta}$$

~~$$\frac{\sin^2 \theta + \cos^2 \theta}{a^2 - b^2}$$~~

div Num & den by $\cos \theta$

$$\rightarrow \frac{\frac{s}{c} - \frac{c}{c} + \frac{1}{c}}{\frac{s}{c} + \frac{c}{c} - \frac{1}{c}} \rightarrow \frac{\tan \theta - 1 + \sec \theta}{\tan \theta + 1 - \sec \theta}$$

$$\rightarrow \frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1}$$

$$\frac{(\sec^2 \theta - \tan^2 \theta)}{a^2 - b^2}$$

HW

#LP: Prove that: $\frac{1}{1+\sin^2 \theta} + \frac{1}{1+\cos^2 \theta} + \frac{1}{1+\sec^2 \theta} + \frac{1}{1+\csc^2 \theta} = 2$

$\swarrow$ $\swarrow$ $\swarrow$ $\swarrow$

$\frac{1}{c^2}$ $\frac{1}{s^2}$

THANK YOU COODIESS !!

